

Linus Kim

| 512-299-8254 | lkim23@utexas.edu | [linkedin.com/in/linuskim2000](https://www.linkedin.com/in/linuskim2000) |

EDUCATION

The University of Texas at Austin, Cockrell school of Engineering	Austin, TX
Master of Science in Engineering - Mechanical Engineering; Robotics	May 2027
Bachelor of Science - Mechanical Engineering; Robotics & Mechatronics	Aug 2019 – May 2025

RESEARCH EXPERIENCES

Human Centered Robotics Lab	Jan 2025 – present
Graduate Research Assistant	Austin, TX

- In-house, end-to-end development of the DRACO 4 humanoid. Designing and manufacturing the entire lower body with CAD, fabricating and assembling mechanical components and integrating electronic hardware systems.
- Developed a ROS2/EtherCAT communication interface utilizing SOEM (Simple Open EtherCAT Master) library to establish real-time communication between control computer and robot hardware.
- Conducted research on learning-based morphology optimization of a humanoid, focusing on oblique lower body axis configurations through PyTorch implementations in Mujoco and Isaac Lab simulation environments.

Hutter Research Group	Aug 2023 – May 2025
Undergraduate Research Assistant	Austin, TX

- Designed a multi-sensor at-home medical device for COPD patients that measures and records flowrate, CO2 concentration, and heart rate from their breathings using Solidworks
- Researched air flow sensor technologies to analyze their applicability to the prototype
- Analyzed 50+ datasets using PCA and developed Python-based UI for data collection and visualization
- Co-authored the presentation poster for the 2024 Defense TechConnect Innovation Summit and Expo

Rehabilitation and Neuromuscular Robotics Lab	June 2024 – Aug 2024
Undergraduate Research Assistant	Austin, TX

- Fabricated and assembled a pair of cable pulley robotic fingers, improving structural integrity and joint play
- Developed 2 prototype tensioner mechanisms that maintained the string system's reliability during experiments
- Assisted research on sensorless force estimation by configuring and calibrating GL80 KV30 actuators

PROJECTS

Basketball Robot	Aug 2025 – Dec 2025
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- Simulated, prototyped, and demonstrated an 8-bar pick and place mechanism integrated with an end effector using the Newton-Raphson method and MATLAB/Python at a live event with 100+ participants

Vision Based Agent Detection System	Oct 2024 – Dec 2024
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- Developed a vision based multi agent control system using OpenCV and Dlib to perform real-time eye tracking
- Implemented SVM classifier with HOG featured detection to process camera input using FilterPy and NumPy
- Created an interactive python application integrating multi-threading and Tkinter/Turtle graphics

Battle Bot	Aug 2024 – Dec 2024
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- Led end-to-end design process including project planning, requirements analysis, preliminary performance calculations and formal design reviews for a battle bot with optimized competitive advantage
- Designed and validated mechanical components through FEA, dynamic load analysis, and iterative prototyping
- Fabricated a robust yet compact prototype through 3D printing, machining, and electrical assembly

Pick and Place Mechanism	Mar 2024 – Apr 2024
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- Simulated, prototyped, and demonstrated an 8-bar pick and place mechanism integrated with an end effector using the Newton-Raphson method and MATLAB/Python at a live event with 100+ participants

SKILLS

- Software: Python, Mujoco, ROS2, MATLAB, SolidWorks
- Hardware: 3D Printing, Machining, CNC Machining, Soldering, Laser Cutting